

Hitesh Rasineni

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ABOUT ME

CS undergraduate working at the intersection of Machine Learning, statistics, and High Energy Physics, with experience in density estimation and clustering for particle-physics data. Focused on advancing density estimation techniques and applying them to Beyond the Standard Model (BSM) physics.

EDUCATION

VIT-AP University, Amaravati, Andhra Pradesh, India

September 2023 – September 2027

B.Tech, CSE (AI & ML)

CGPA: 8.05 / 10

Sree Vidyanikethan International School, Tirupati, India

2023

CBSE (10+2)

Grade: 91.6 / 100

RESEARCH

Leptonic Mono-Z Dark Matter Search with Neural Spline Flows

CMS Run 2015D Open Data

Code: github.com/HiteshRasineni/Leptonic-Mono-z-CMS2015-DarkMatter-Search

- Trained five Neural Spline Flows on 2.32 fb^{-1} of CMS Run 2015D data (parallel $\mu\mu/ee$ channels; 40 observables from MINIAOD reduced to a 37-dim feature vector): two channel-specific SM background flows learned from control-region Drell–Yan events, and three mediator-specific signal flows (vector, axial-vector, scalar) from MadGraph-based MC.
- Used the per-event log-likelihood ratio $\log p(x|\text{DM}) - \log p(x|\text{SM})$ as the test statistic, followed by a simultaneous SR+VR binned profile-likelihood fit; set 95% CL observed limits on the signal strength of $\mu < 0.018$ (scalar), 0.036 (vector), and 0.050 (axial-vector), attributing the excess over expectation to a high- E_T^{miss} background-modelling residual rather than a DM signal.

Hadronic Mono-Z Dark Matter Sensitivity with Flow Matching

CMS Run 2015D HTMHT Open Data

Code: github.com/HiteshRasineni/CMS2015DarkMatterSearch-HTMHT

- Built the full simulation-to-inference pipeline (MadGraph5_aMC@NLO $\rightarrow$ Pythia 8 $\rightarrow$ Delphes signal generation; feature extraction from 20.7M raw events, 1.44M selected across 5 unprescaled HLT paths) and modeled the background density with a conditional flow-matching normalizing flow trained on real HTMHT events.
- Applied sentinel imputation for missing-object features, held-out NLL scoring, an offline signal trigger proxy, and a $B \geq 20$ yield floor to suppress low-statistics argmax bias; projected expected significances of 2.89σ , 7.62σ , and 7.41σ for three simplified-model benchmarks, with ablations showing extra-jet kinematics carry 53–71% of the discriminating power.

Improving Discovery-Significance Stability in Higgs Event Classification

HiggsML $H \rightarrow \tau^+\tau^-$

benchmark

with J.J. Pujari, P.A. Immadi, T. Bikku, R.S. Puppali (VIT-AP University / Amrita).

- Developed a supervised contrastive pre-training framework combined with a parallel FT-Transformer + XGBoost ensemble; repeated 5×5 -fold cross-validation showed reduced fold-to-fold variance in Approximate Median Significance (AMS = 3.74 on the full dataset) versus focal-loss training, enabling more stable threshold selection. Published in *Discover Artificial Intelligence* (Springer Nature).

SELECTED PROJECTS

The Unlikely Lab — Personal Research Publication Platform

hiteshrasineni.github.io/the-unlikely-lab

Code: github.com/HiteshRasineni/the-unlikely-lab

- Designed and deployed a full research-publication platform that hosts complete, typeset HTML versions of arXiv-style physics papers (sections, numbered equations, figures, tables, bibliography) alongside an MDX-driven content system for research notes and project pages.
- Built an automated LaTeX-to-HTML paper build pipeline with Pandoc, including MathML/KaTeX rendering, figure/table validation reports, and per-paper static generation; the site is a statically exported Next.js (React, TypeScript, Tailwind CSS) application deployed via GitHub Actions CI/CD to GitHub Pages.

PUBLICATIONS

Dark Matter Mono-Z Analysis

- **Mono-Z Dark Matter Search with Neural Spline Flows Using CMS Run 2015D Open Data.**
H. Rasineni, B. Chebrolu. arXiv preprint, 2026 (submitted for journal publication). *Leptonic decay channel*.
DOI: 10.48550/arXiv.2607.13771
- **Hadronic Mono-Z Dark Matter Sensitivity with Flow Matching on CMS Open Data.**
H. Rasineni, B. Chebrolu. Preprint (under review), Research Square, 2026. *Hadronic decay channel*.
DOI: 10.21203/rs.3.rs-10634384/v1

Improving Stability of Discovery Significance in Higgs Boson Event Classification using Contrastive Representation Learning.

J.J. Pujari, P.A. Immadi, H. Rasineni, T. Bikku, R.S. Puppala. *Discover Artificial Intelligence* (Springer Nature), 2026.

DOI: 10.1007/s44163-026-01683-5

TECHNICAL SKILLS

Languages: Python, Java, SQL, JavaScript (React)

ML/Data: PyTorch, scikit-learn, OpenCV, Pandas, Matplotlib, Density Estimation (Normalizing Flows, Likelihood Ratios)

Tools: Git, Docker, Google Colab, uproot, Latex

CERTIFICATIONS

- **Microsoft Certified: Azure AI Fundamentals** (Credential)
- **AWS Academy Graduate – Cloud Foundations (Training Badge)** (Credential)
- **WorldQuant University, Applied AI Lab – Deep Learning for Computer Vision** (YOLOv8, MTCNN, Inception-ResNetV1) (Credential)
- **IBM Machine Learning Specialist (Professional V1)** (EDA, ensembles, transfer learning) (Credential)
- **NVIDIA DLI – Fundamentals of Deep Learning** (Credential)
- **NVIDIA DLI – Transformer-based NLP** (Credential)